

Plenitude Principles and Tolerance Puzzles

Advanced Introduction to Metaphysics

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1. How semantic plasticity defeats the security argument for Non-Contingency

Consider a *de re* Tolerance Argument, e.g. where Tolerance is ‘Every collection of atoms that chemically matches Woody’s originating atoms and overlaps them by at least 90% is one from which Woody have originated’.

Independence If Tolerance could easily have been false, we could easily have falsely believed it.

Security If Tolerance is true, we couldn’t easily have falsely believed it.

Easiness If Tolerance could have been false, it could easily have been false.

Non-Contingency If Tolerance is true, it couldn’t have been false.

Independence is motivated by the thought that if Tolerance could easily have been false, it could easily have been false while we still went around uttering the same *sentences*.

But what if, in those worlds that, we say something different—and something true—in uttering the sentences that in the actual world we use to say that Tolerance is true?

The obvious way of fleshing this out: our uses of ‘Woody’ (or ‘this table’) in those near-by worlds where Woody isn’t tolerant don’t refer to Woody, but to something tolerant that coincides (there) with Woody.

2. Plenitude principles: rough idea

That response to Tolerance Puzzles generates an urgent demand for some general principle that predicts the existence of the required multiplicity of coinciding objects whose modal properties differ in the right ways and not in unwanted ways.

Fortunately there is a range of defensible Plenitude principles, which in one way or other flesh out the idea that every ordinary object coincides with *enormously many* others, which between them cover a whole spectrum of modal profiles.

In addition to defusing the Security Argument, these principles also defuse the concern that the contingent truth of Tolerance would make the actual world objectionably ‘special’.

3. Other ways of arguing for Plenitude principles

- Charity to odd counterfactual communities
- Charity to variability in our own dispositions across contexts

- “Abductive” inference needed at some point to get from examples to any general Plenitude principle.

4. Some more precise Plenitude principles

Location Plenitude: For any property F , there is an object x such that necessarily, for any spacetime point y , x is located at y if and only if y is F .

- Possible modifications: require F to be instantiated; require F to be possibly instantiated; require F to demarcate a topologically well-behaved region; require F to entail being “matter-filled”; ...
- Note that the principle doesn’t say ‘exactly one’. However, if one thinks that there are cases of necessary co-location (as in Fine’s “Prittle and Prattle” example), one will be in the market for a stronger principle that predicts this.

If we define ‘coincident’ as ‘located at the same spacetime points’, Location Plenitude implies:

Coincidence Plenitude For any property F , there is an object x such that it is metaphysically necessary that for any object y , y coincides with x iff something is F and y coincides with everything F .

- Or equivalently: *Every undiscriminating, unrepeatable property is tracked by an object*, where F is undiscriminating := necessarily anything that coincides with something F is F ; F is unrepeatable := necessarily any two F things coincide; x tracks F := necessarily any y coincides with x iff y is F .

Many principles of a Plenitudinous sort in the literature are stated using *mereological* vocabulary. Here’s a relatively non-committal such principle, with the same “two level” form as Location Plenitude.

Fundamental Part Plenitude: For any property F , there is an object x such that it is metaphysically necessary that for any fundamental object y , y is part of x if and only if y is F .

Here’s a “one-level” principle (see Hovda, ‘Tensed Mereology’):

Fusion Plenitude: For every property F , there is an object x such that necessarily: every F object is part of x and every part of x shares a part with some F object.

- An important choice point for Plenitude-lovers (especially those who theorise in mereological terms): do familiar objects have parts with crazy modal profiles, e.g. do I have a part that would have coincided with my bike if it had been raining today?

- Saying ‘no’ might push one towards a ‘hierarchical’ Plenitude principle where the realm of objects is reminiscent of the realm of sets as depicted in ZFC. One must then be careful to avoid principles that succumb to Russell’s paradox, like this one:

Immediate Part Plenitude: For any property F , there is an object x such that necessarily, for any object y , y is an *immediate part* of x iff y is F .

Some but of the above Plenitude principles naturally suggest strengthenings involving higher order identity rather than modality:

Identity-Strength Location Plenitude: For any property F , there is an object x such that to be a spacetime point at which x is located is to be an F spacetime point.

5. Plenitude and common nouns

Leslie’s (apparent) view: every axe coincides with many other axes.

- Worry 1: this just gives up on quantified Tolerance speeches (‘Every table [in the room] could have been made of somewhat different matter’...)
- Worry 2: this seems to lead to rampant error in counting judgments.
- A possible response to Worry 2: counting sentences aren’t connected to identity sentences in the orthodox way.
- A possible response to both worries: contextual quantifier domain restriction is pervasive, and ordinary domains only contain one of any collection of coinciding tables.

Yablo’s view: necessarily, every ship is tolerant; but no ship is tolerantly a ship.

- Worry: this conflicts with lots of ordinary judgments.

Our view: words like ‘table’, ‘axe’, ‘ship’ are subject to semantic plasticity too.

- So, externalism is more pervasive (and ‘Non-Twin-Earthability’ rarer) than some think.
- Also, *temporal* externalism is true: worlds whose physical histories are the same up to t can differ semantically before t .

6. Some choice points

7. The challenge from speech reports

8. The challenge from people

